

MATH 345 Skeleton Notes

Unit 5: Optimization and training

Section 5.2: Directional derivatives and gradient descent

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Directional derivatives

Activity

Find the system of equations satisfied by the critical points of the function

$$f(x, y, z) = x^2 e^{yz} + y \sin(xz) + z \cos(xy).$$

Using the calculus we have developed, it is usually straightforward to write the equations that critical points must satisfy. The activity also shows the problem: those equations may be nonlinear and difficult or impossible to solve exactly. This is different from the linear systems in Unit 2, where Gaussian elimination gave a systematic exact method. For such problems, one often uses numerical methods. Gradient descent starts with an initial guess $\mathbf{x}_0$ and produces iterates

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The positive scalar α is the step size. The minus sign moves in a direction where f initially decreases. The method does not automatically converge; the function, starting point, and step size matter. To discuss this further, we must introduce more general kinds of derivatives. Recall that the partial derivatives of a function f are defined as

$$\frac{\partial f}{\partial x_i}(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{e}_i) - f(\mathbf{a})}{h}.$$

By replacing $\mathbf{e}_i$ by a more general unit vector, we obtain the concept of a *directional derivative*.

Definition Let f be a scalar-valued function. Then if $\mathbf{v}$ is a unit vector, the

directional derivative of f in the direction of the vector $\mathbf{v}$, denoted $D_{\mathbf{v}}f$, is defined as

$$D_{\mathbf{v}}f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{v}) - f(\mathbf{a})}{h},$$

provided the limit exists.

The directional derivative represents the rate of change of the function f at the point $\mathbf{a}$ in the direction of the unit vector $\mathbf{v}$. For a function $f(x, y)$ of two variables, the directional derivative $D_{\mathbf{v}}f(\mathbf{a})$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x, y)$ with the plane containing both the line passing through $\mathbf{a}$ with direction vector $\mathbf{v}$, and the z axis.

Remark Partial derivatives are a special case of directional derivatives, i.e. we have

$$\frac{\partial f}{\partial x_i} = D_{\mathbf{e}_i}f.$$

Despite partial derivatives being a special case of directional derivatives, in most circumstances we can compute directional derivatives using

partial derivatives, or equivalently, the gradient.

Theorem Suppose f is a function differentiable at a point $\mathbf{a}$. Then for each unit vector $\mathbf{v}$, $D_{\mathbf{v}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

Proof We apply the chain rule. We note that $D_{\mathbf{v}}f(\mathbf{a})$ is equal to the derivative of the function $g(t) = f(\mathbf{a} + t\mathbf{v})$ at $t = 0$ (both derivatives are defined as the same limits). If we write $\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}$, then $g = f \circ \mathbf{c}$. But the chain rule (Chain Rule for One Independent Variable) implies that

$$g'(0) = (\nabla f)(\mathbf{c}(0)) \cdot \mathbf{c}'(0) = (\nabla f)(\mathbf{a}) \cdot \mathbf{v}.$$

The theorem greatly simplifies the computations of directional derivatives.

Activity: Dot product controls direction

Suppose

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Compute $D_{\mathbf{v}}f(\mathbf{a})$ for each unit vector below:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_4 = -\frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

1. Which listed direction gives the largest directional derivative?
2. Which listed direction gives the smallest directional derivative?
3. How is this the Unit 1 dot product appearing again?

Tags. [U5-L02, U1-L01 | C+R | Core]

Activity

Let $f(x, y) = x^2y + y^2$.

Compute the directional derivative $D_{\mathbf{v}}f(1, 2)$, where $\mathbf{v} = (1/\sqrt{2}, 1/\sqrt{2})$, and $f(x, y) = x^2y + y^2$.

Activity

Let $f(x, y) = x^2y + y^2$.

Find the directional derivative of f at the point $(-1, 3)$ in the direction of the vector $(-1, \sqrt{3})$. That is, compute $D_{\mathbf{v}}f(-1, 3)$, where $\mathbf{v}$ is the unit vector in the direction of $(-1, \sqrt{3})$.

Activity

Use the gradient of the function $f(x, y) = x^3y$ to find the directional derivative of f at $(2, 1)$ in the direction of the unit vector which we placed at $(2, 1)$ points in the direction of the point $(3, 5)$.

Activity

Let $f(x, y) = x^2 e^y$. Find the directional derivative of f at $(2, 0)$ in the direction pointing from $(2, 0)$ to $(3, 0)$.

Theorem Suppose that a function f is differentiable at a point $\mathbf{a}$.

- If $\nabla f(\mathbf{a}) = 0$, then $D_{\mathbf{v}}f(\mathbf{a}) = 0$ for any unit vector $\mathbf{v}$.
- If $\nabla f(\mathbf{a}) \neq 0$, then $D_{\mathbf{v}}f(\mathbf{a})$ is maximized, among unit vectors $\mathbf{v}$, when

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

It is minimized when

$$\mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

The maximum value is $\|\nabla f(\mathbf{a})\|$, and the minimum value is $-\|\nabla f(\mathbf{a})\|$.

Proof We have already verified the theorem when $\nabla f(\mathbf{a}) = \mathbf{0}$ in the theorem. When $\nabla f(\mathbf{a}) \neq \mathbf{0}$, the theorem and the Cauchy-Schwarz Inequality The Cauchy-Schwarz inequality says that

$$|D_{\mathbf{u}}f(\mathbf{a})| = |\nabla f(\mathbf{a}) \cdot \mathbf{u}| \leq \|\nabla f(\mathbf{a})\| \|\mathbf{u}\| = \|\nabla f(\mathbf{a})\|,$$

where the inequality is an *equality* precisely when $\mathbf{u}$ either points in the direction of $\nabla f(\mathbf{a})$, or points in the opposite of this direction. But when $\mathbf{u}$ points in the direction of $\nabla f(\mathbf{a})$, i.e. when $\mathbf{u} = \nabla f(\mathbf{a})/\|\nabla f(\mathbf{a})\|$, we have

$$\begin{aligned} D_{\mathbf{u}}f(\mathbf{a}) &= \nabla f(\mathbf{a}) \cdot \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|} \\ &= \frac{\nabla f(\mathbf{a}) \cdot \nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|} = \frac{\|\nabla f(\mathbf{a})\|^2}{\|\nabla f(\mathbf{a})\|} = \|\nabla f(\mathbf{a})\|. \end{aligned}$$

Conversely, one can check that when $\mathbf{u} = -\nabla f(\mathbf{a})/\|\nabla f(\mathbf{a})\|$, then

$$D_{\mathbf{u}}f(\mathbf{a}) = -\|\nabla f(\mathbf{a})\|.$$

So we see that these two vectors are precisely the maximizer and minimizer of the directional derivatives.

Activity

Find the direction for which the directional derivative of $f(x, y) = 3x^2 - 4xy + 2y^2$ at $(-2, 3)$ is a maximum. What is the maximum value of the directional derivative?

Gradient descent

Remark We now return to the gradient descent algorithm, a method for trying to approximate the local minima of a function f . The algorithm works as follows:

1. Start at an initial point $\mathbf{x}_0$.
2. Compute the gradient $\nabla f(\mathbf{x}_0)$.
3. Move in the direction of the negative gradient: set $\mathbf{x}_1 = \mathbf{x}_0 - \alpha_0 \nabla f(\mathbf{x}_0)$, where $\alpha_0 > 0$ is a step size (or learning rate) parameter.
4. Repeat steps 2-3 with step sizes $\alpha_1, \alpha_2, \dots$ to successively define points $\mathbf{x}_2 = \mathbf{x}_1 - \alpha_1 \nabla f(\mathbf{x}_1)$, $\mathbf{x}_3 = \mathbf{x}_2 - \alpha_2 \nabla f(\mathbf{x}_2)$, and so on, with the hope that these points converge to a local minimum.

Intuitively, gradient descent works because each successive value $\mathbf{x}_k$ is obtained by shifting the previous value $\mathbf{x}_{k-1}$ in the direction in which the function f decreases the fastest. The algorithm is likely to find local minima of a function f , although a rigorous analysis of this algorithm is quite involved.

Activity: Reading one gradient descent update

Assume $\text{grad_f}(x)$ computes $\nabla f(x)$. Consider the line of code

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x = x - alpha * grad_f(x)
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1. What mathematical update rule does this line represent?
2. Which quantity is the learning rate?
3. Why must $\text{grad_f}(x)$ have the same shape as x ?
4. What would change if the minus sign were a plus sign?

Tags. [U5-L03 | C+T | Core]

Activity

Find an approximate minimum of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$$

using gradient descent. Use the starting point $\mathbf{x}_0 = (0, 0)$, a constant step size of 0.1, and stop after computing $\mathbf{x}_3$. You may use a calculator to perform these computations.

Remark This algorithm can be rather cumbersome to compute by hand, but if the gradient of a function is easily determined, can be easily done to a large number of iterations on a computer.

Activity

Calculate $D_{\mathbf{u}}f(1, -2, 3)$ in the direction of $\mathbf{v} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ for the function

$$f(x, y, z) = 5x^2 - 2xy + y^2 - 4yz + z^2 + 3xz.$$